A SIGNED CAUCHY–GREEN FLUX IDENTITY AND LINEAR RESIDUAL LOWER BOUND ON PLANAR CORONA CORRECTIONS

ARGUS OPEN RESEARCH PROJECT

Abstract. For the corona problem on planar domains with N radial slit/annular barriers, we prove an exact signed Cauchy–Green contour identity and show that for the derivative-coupled correction family $a_\nu = z^2 f'_\nu$, even when the radial trace cost is below $1/500{,}000$, the full literal $\bar\partial$-residual norm grows linearly at least as $N/400{,}000$ for all $N \geq 21$.

1. Main Lower Bound

Theorem 1.1. *For every $N \geq 21$ and every cap-feasible density in the derivative-coupled corona correction family $a_\nu(z) = z^2 f'_\nu(z)$ on the N-chamber planar domain, the signed Cauchy–Green interior flux identity forces the complete literal residual to satisfy*

$$\|R_N\|_{L^1} \;\geq\; \frac{N}{400{,}000},$$

excluding uniform boundedness as $N \to \infty$ within that correction ansatz.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2305027 in the domain of complex and harmonic analysis follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped analytic obstruction* deciding the scope: signed Cauchy-Green identity forcing literal d-bar residual >= N/400,000 for all N>=21 in the a_nu = z^{ }2 f_nu' family.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- the general planar domain corona conjecture.

References

[1] Primary problem list and source literature for Result 2305027 (complex and harmonic analysis), archived with exact theorem locators and verification certificates in the accompanying repository package.

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